

ADAPTIVEFUSION: Low Power Scene Reconstruction

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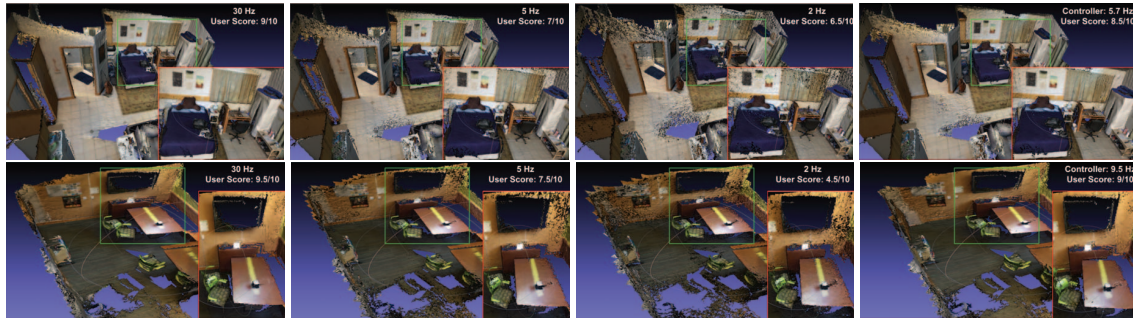


Figure 1: Meshes generated with various constant and dynamic scene reconstruction frequencies for two different scenes.

ABSTRACT

Towards deploying scene reconstruction on mobile devices with a stringent power budget, we propose ADAPTIVEFUSION to meet the power envelope by dynamically changing the reconstruction frequency while maintaining acceptable quality. Compared to a standard 30 Hz baseline, ADAPTIVEFUSION reduces frequency, and thus power, by $4.9\times$ on average and is applicable to other fusion-based algorithms.

Index Terms: Human-centered computing—Mixed/augmented reality—Human-centered computing—Virtual reality

1 INTRODUCTION

Scene reconstruction creates a 3D model of the user’s surroundings to allow interaction between virtual and real worlds. As a key technology to XR, it has flourished in the past decade [6, 7], but few have studied it in the resource-constrained environment of XR systems. Despite the power budget for XR systems being limited [3], most scene reconstruction algorithms [4, 9] are running at camera frequency (30 Hz) and developed on powerful systems. Methods that can run on mobile devices [4] still consume power beyond the budget of XR devices.

This paper addresses the problem of decreasing power consumption for real-time scene reconstruction on power-constrained XR devices. Different from previous works [8, 9] that optimize the amount of work done in processing each frame, we reduce the frequency of the data fusion process to mitigate power consumption.

Fig. 1 compares the meshes obtained by running depth fusion at different frequencies for two different sequences. It shows that while there is an opportunity to reduce power by running fusion at a frequency lower than 30 Hz mesh, the exact frequency varies from sequence to sequence. Always choosing a constant high frequency would unnecessarily increase power consumption, whereas choosing a low frequency would degrade mesh quality in some sequences.

We propose ADAPTIVEFUSION, a dynamic controller that adjusts the frequency of fusion over time to achieve a balance between

power consumption and mesh quality. ADAPTIVEFUSION reduces fusion frequency, and thus power, by $4.9\times$ on average, while still giving acceptable quality meshes, compared to the state-of-the-art scene reconstruction implementation.

2 ADAPTIVEFUSION

Scene reconstruction is typically run at a constant frequency of 30 Hz to suit the high frequency needed for the ICP-based pose trackers. However, in a system with a pose estimator that is external to scene reconstruction, such restriction no longer applies, and voxel fusion—referred to simply as fusion from here on out—can potentially be performed at a lower frequency, reducing system power and hardware utilization.

However, as shown in Fig. 1, the main challenge with choosing a lower frequency is that the optimal fusion frequency depends on both the scene and the user’s motion. Thus, determining an optimal, constant frequency *a priori* is not feasible.

We propose an online controller for real-time scene reconstruction that dynamically chooses the fusion frequency based on the scene and user’s motion, as opposed to a constant 30 Hz as seen in existing systems. We assume a voxel-based algorithm with an external pose tracker. However, ideas presented in this paper are broadly applicable and not limited to a specific implementation.

Our design is inspired by [8]¹ but differs in two salient ways. First, we monitor the voxel content of the scene, a fundamental metric that directly reflects the necessity of fusion to determine the instantaneous frequency; i.e., the inverse of the waiting time until the next frame is fused. Second, we output a fusion frequency that is applicable to various fusion-based scene reconstruction algorithms, avoiding manual efforts to design control knobs.

3 EXPERIMENTAL SETUP

Implementation We implement our system in the open-source InfiniTAM v3 framework [4]. InfiniTAM provides separate modular components for tracking and fusion. Since we focus on scene reconstruction, to isolate reconstruction errors from pose estimation errors, we inject ground truth poses provided by the data set to InfiniTAM instead of estimating poses using its tracking module. Using ground truth poses does not diminish the evaluation results, as our design

¹ [8] tracks pose distance to adjust handcrafted algorithm-specific knobs.

Table 1: ScanNet sequences used to tune the controller parameters. Each sequence consists of depth and color images both at 30 Hz.

Sequence	scene0000	scene0002	scene0004	scene0005	scene0010	scene0101
Length	5578	5193	929	1159	2514	1408

Table 2: ScanNet sequences used in the testing set.

Sequence	scene0054	scene0233	scene0331	scene0465	scene0613
Length	6629	8268	6030	6307	6001

only affects the frames selected for reconstruction and is orthogonal to how poses are generated.

For frequency reduction and the online controller, we make no changes to the core algorithm, and simply control the input to the pipeline. We assume the depth camera always runs at the frame rate of the performance baseline InfiniTAM. In optimized configurations where depth fusion is run at a frequency reduced by a factor X , our system will invoke fusion only for every X th frame, saving energy on the $X - 1$ previous frames.

Sequences We use sequences from ScanNet [1] to evaluate our system, listed in Table 1 and Table 2. The sequences cover a variety of scenes, camera trajectories, and difficulties, and range in length from 929 to 8268 frames (31 seconds to 4.5 minutes), reflecting typical execution periods of scene reconstruction use cases. We manually tune the controller using a training set consisting of the ScanNet sequences listed in Table 1, and use the remaining sequences as the testing set.

Metrics To evaluate the quality of the reconstructed scenes, we use F-score as defined by [5]. We extract a mesh from the final voxel volume using marching cubes, and post-process it by removing redundant vertices. We obtain ground truth meshes by passing the ground truth poses of each sequence to InfiniTAM at 30 Hz. We assume that power consumption scales linearly with frequency.

Configurations We use the following frequencies to evaluate each sequence: 30 Hz, 15 Hz, 10 Hz, 7.5 Hz, 5 Hz, 2 Hz, and ADAPTIVEFUSION. We use 5 mm voxels, a truncation band of 20 mm, and a viewing frustum of [0.3, 4] m in all cases.

4 RECONSTRUCTION EVALUATION

Quantitative Results Fig. 2 and 3 show the F-score of each constant frequency configuration and ADAPTIVEFUSION for the ScanNet training set and test set, respectively. Each figure shows the F-score with both a 2 cm threshold and a 5 cm threshold (representing approximately 0.4% and 1% of the scene’s scale). The F-scores are computed by taking the 30 Hz mesh as the reference. We cap the range of F-score values on the y-axis for readability, and we use a log scale on the x-axis to spread the data points evenly. The **star marker** denotes the F-score and average frequency of ADAPTIVEFUSION in a sequence. We make the following observations.

First, for a given target F-score, the minimum frequency that meets that score varies from sequence to sequence. For instance, the acceptable frequency range is 4 - 16 Hz if an F-score of 0.98 (5 cm threshold) is chosen for the ScanNet training set in Fig. 2.

Second, ADAPTIVEFUSION (denoted using star markers) is able to achieve a good trade-off between mesh quality and frequency, selecting unique frequencies for each sequence and generating meshes with acceptable quality. In some sequences, the star marker is even higher than the constant frequency trend line, meaning that the controller can either achieve similar mesh quality by fusing fewer frames or fuse the same number of frames and produce a better mesh. We find that these results hold with both error thresholds.

Finally, ADAPTIVEFUSION shows generalizability by producing good meshes in the test sequences; e.g., the star markers are above the constant frequency trend line for some sequences in Fig. 3.

Qualitative Results F-scores measure the relationship between mesh quality and frequency but do not indicate perceptual quality.

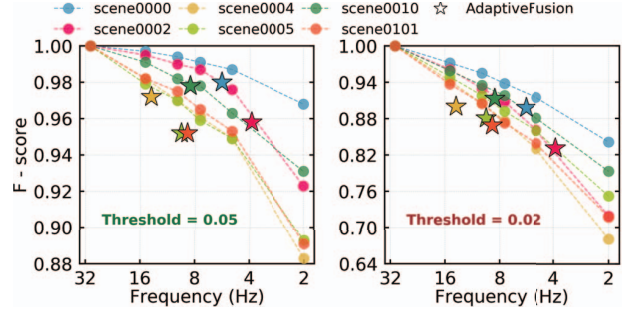


Figure 2: F-scores with 5cm/2cm thresholds for the training set.

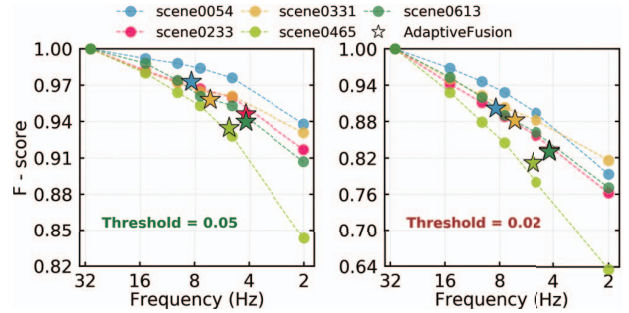


Figure 3: F-scores with 5cm/2cm thresholds for the testing set.

Selecting a target F-score is difficult as the score changes with the error threshold. Therefore, we also visually inspect all the meshes.

Fig. 1 shows the reconstructed meshes for two sequences from ScanNet, each with four configurations: constant 30 Hz (best achievable mesh), the constant frequency that generates a minimally acceptable mesh, the constant frequency that generates a more acceptable mesh, and ADAPTIVEFUSION. Fig. 1 corroborates the F-score results, with both sequences exhibiting different optimal frequencies, and ADAPTIVEFUSION successfully adapting to each scene. In both cases, ADAPTIVEFUSION generates a mesh that is better than the minimally acceptable mesh and comparable to the more acceptable mesh. More detailed results are available in [2].

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